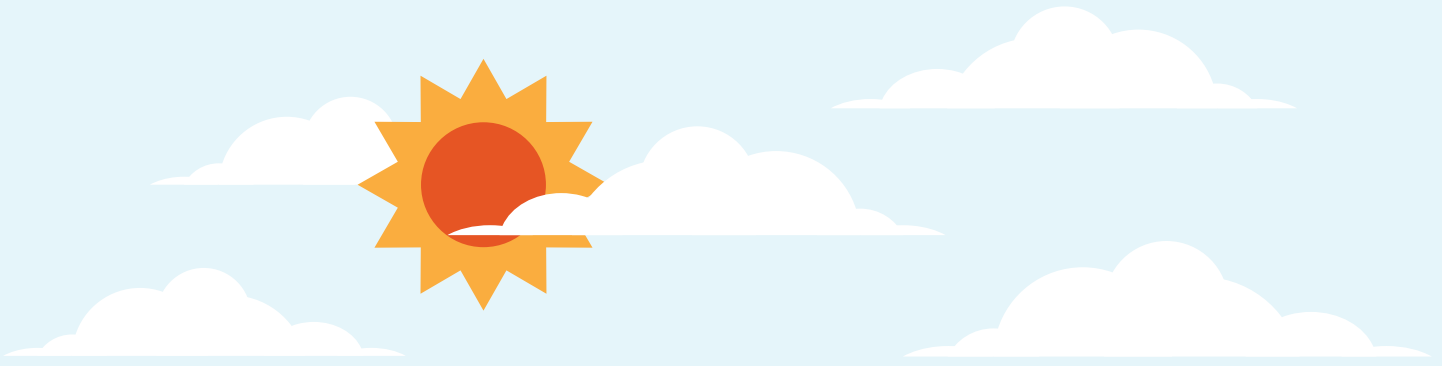
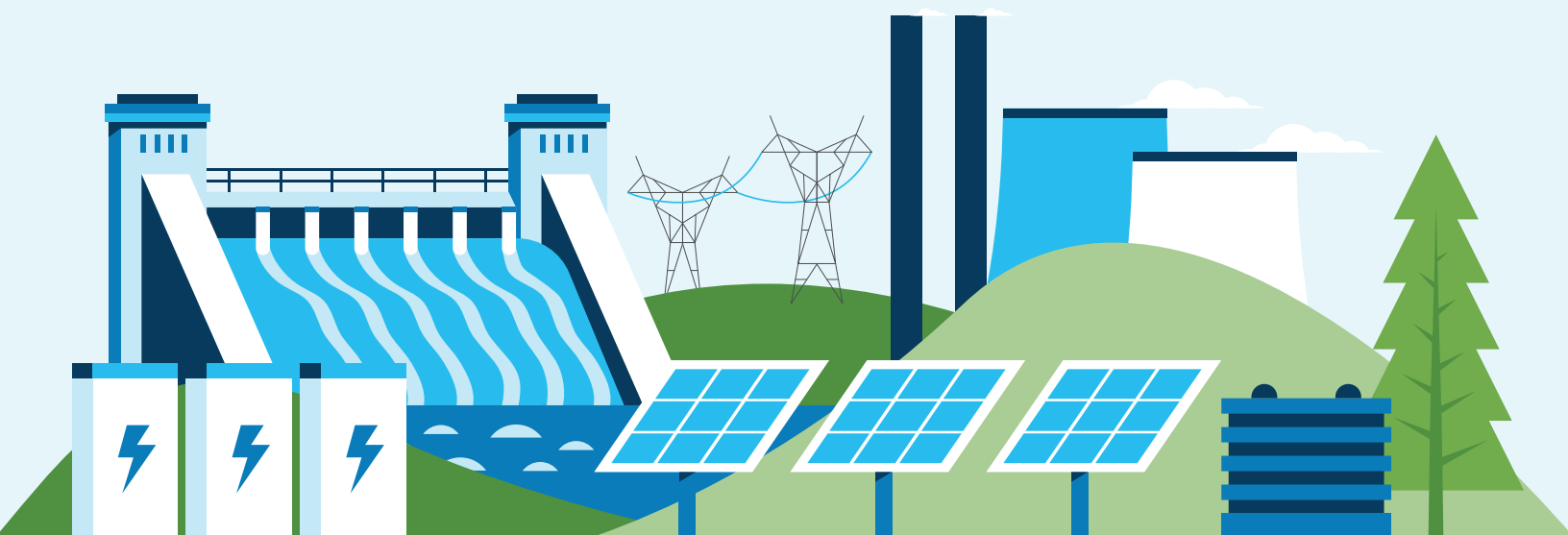




TRANSMISSION 101

HOW ELECTRICITY GETS TO YOUR HOME



Building the Future of Energy

We are building the future of energy by investing in a stronger and more resilient power grid. As one of the best places in the country to live, work, and do business, Georgia has seen an unprecedented growth in population and industry over the past years.

This boom also comes with much higher demand for electricity and makes our commitment to supplying our growing state with clean, safe, reliable, and affordable energy more important. One crucial part of us building the future of energy is updating and building new transmission infrastructure.

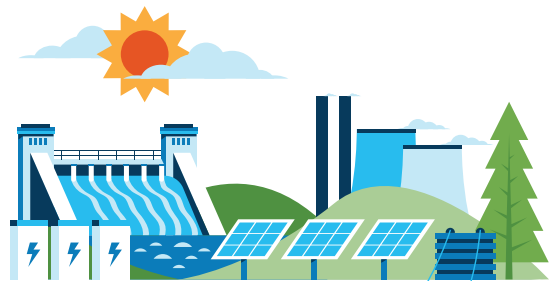
Reliable, Resilient, and Forward-looking Energy

Transmission lines and substations are the backbone of a reliable, resilient, and forward-looking energy grid. They ensure electricity flows efficiently from generation sources to homes and businesses, support economic growth by meeting rising demand, and enable the integration of cleaner energy. By upgrading aging infrastructure and building new connections, we're not only strengthening the grid but also preparing Georgia for a more electrified future. Just as important, we recognize that these projects touch the lives of our communities. That's why we're committed to open communication, thoughtful planning, and minimal disruption—so we can build the future of energy together.

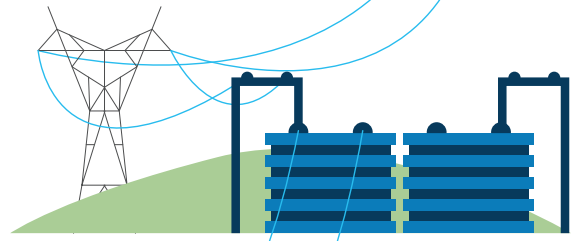
The Journey of Electricity

Electricity is generated at power plants using various energy sources like coal, natural gas, nuclear power, or renewables. It then travels through high-voltage transmission lines to substations, where the voltage is reduced for distribution. Finally, electricity is delivered to homes and businesses through lower-voltage distribution lines.

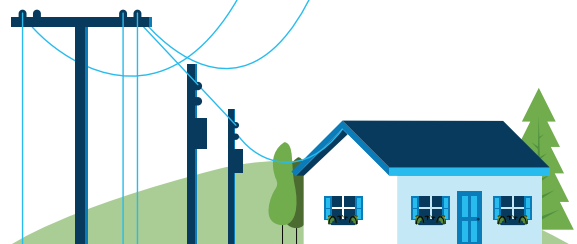
GENERATING SOURCES



TRANSMISSION LINES



SUBSTATION



OVERHEAD TRANSFORMERS



DISTRIBUTION LINES

UNDERGROUND TRANSFORMERS

Why is the Grid Important?

RELIABILITY:

Keeps power flowing during high demand or outages.

Transmission lines are crucial for maintaining the stability and reliability of the power grid in our state. They ensure that electricity can be transported efficiently from power plants to consumers, even during periods of high demand or unexpected outages. By connecting different areas, transmission lines allow for the sharing of electricity resources, which helps balance supply and demand across the grid. This interconnectivity is particularly important during extreme weather events or other disruptions, as it enables the grid to remain resilient and continue providing power to homes and businesses.

ECONOMIC GROWTH:

Supports new industries and population growth.

Transmission lines play a vital role in supporting economic growth by enabling the development of new industries and accommodating population growth. As communities expand and new businesses emerge, the demand for electricity increases. Transmission infrastructure ensures that this growing demand can be met by delivering power from diverse sources, including renewable energy. This not only supports the establishment of new industries but also attracts investments and creates jobs.

Why is Building and Updating Transmission Infrastructure in Georgia so Crucial?

AGING INFRASTRUCTURE:

Replacing assets nearing end of life with the latest technology.

GROWING DEMAND:

Electrification of vehicles, homes, and industries.

SEVERE WEATHER RESILIENCE:

Need to withstand extreme weather.

Modernizing the grid ensures reliable power delivery, supports economic growth, and enhances the system's ability to withstand and recover from extreme weather events. Upgrading transmission lines and substations helps reduce the frequency and duration of power outages, providing a more stable and resilient power supply. Expanding the transmission network accommodates the increasing energy needs of Georgia's growing population and industries.

Substations: The Grid's Control Points

Types: **Transmission Substations:** Step up/down voltage. **Distribution Substations:** Feed local distribution networks.

What is a substation?

Substations play a vital role in managing the flow of electricity, allowing for reliable and safe power distribution across the state. One of the main roles of substations is to convert electricity into different voltages (from high to low, or vice versa). This is needed so electricity can be transmitted throughout the state and then distributed throughout local neighborhoods and into our homes, businesses, and buildings. Substations also house equipment like transformers, circuit breakers, and switches to control and protect the electrical grid.

Functions:

Voltage Regulation

Substations use devices like voltage regulators and tap-changing transformers to maintain consistent voltage levels, ensuring reliable electricity delivery and protecting equipment from fluctuations and maintaining power stability

Switching and Routing

Switchgear such as circuit breakers and disconnect switches control electricity flow between circuits, allowing faults to be isolated, maintenance to be performed safely, and outages to be addressed quickly—ensuring efficient power distribution and grid reliability.

Protection and Monitoring

Relays, breakers, and monitoring systems in substations detect and isolate faults, while providing real-time data on voltage levels and current flow to prevent outages and equipment failures.

The Types of Upgrades and Why

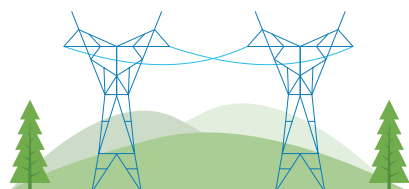
Upgrading Existing Lines

Replacing Aging Equipment:

Replacing aging equipment is essential to prevent failures, reduce maintenance costs, and ensure the continuous and reliable delivery of electricity.

Increasing Capacity:

Upgrading lines with advanced technology boosts capacity, meeting growing demand without building new infrastructure.



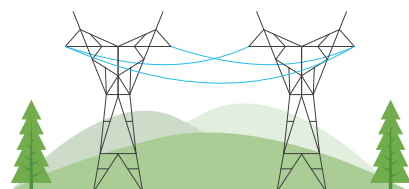
New Transmission Lines

Connecting Remote Renewables:

New transmission lines connect renewable sources to the grid, advancing clean energy initiatives and promoting sustainability.

Enhancing Flow and Reliability:

New transmission lines optimize the electrical system flow, allowing for greater efficiency and reliability of energy distribution.



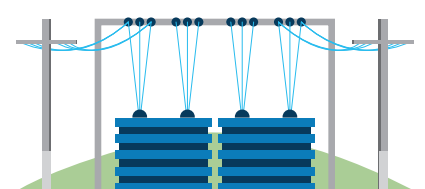
New Substations

Supporting Load Growth:

New substations support growing electricity demand, maintaining grid reliability.

Enhancing Grid Flexibility:

Substations also increase grid flexibility, enabling better power flow, renewable integration, and demand response.



What Are Transmission Lines?

Electric transmission lines are specialized high-voltage lines that carry electricity over long distances, such as from a power station to a city.

Types of lines:

Power travels down a series of different size transmission lines: 500 kilovolt, 230 kV, 115 kV and some 69 kV and 46 kV.

Components:

Towers:

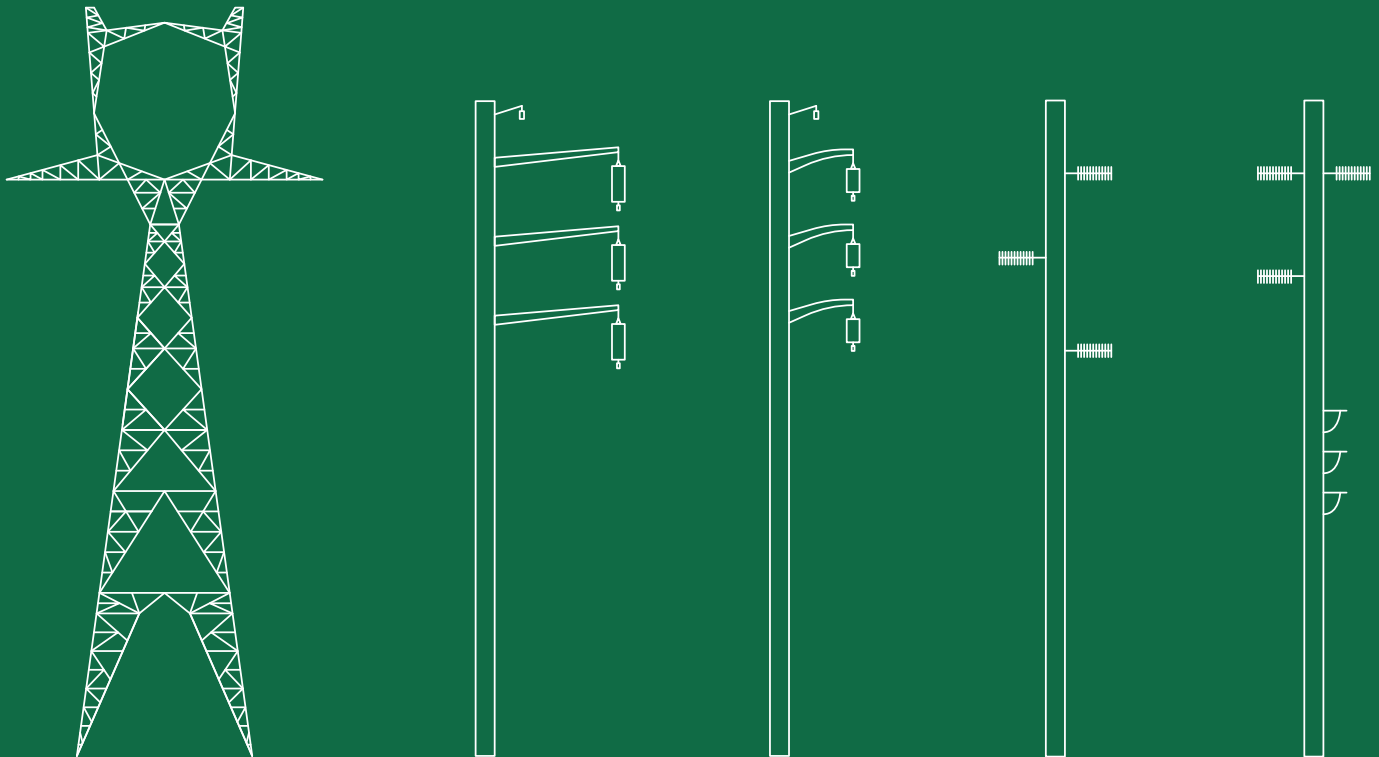
Support overhead lines.

Conductors:

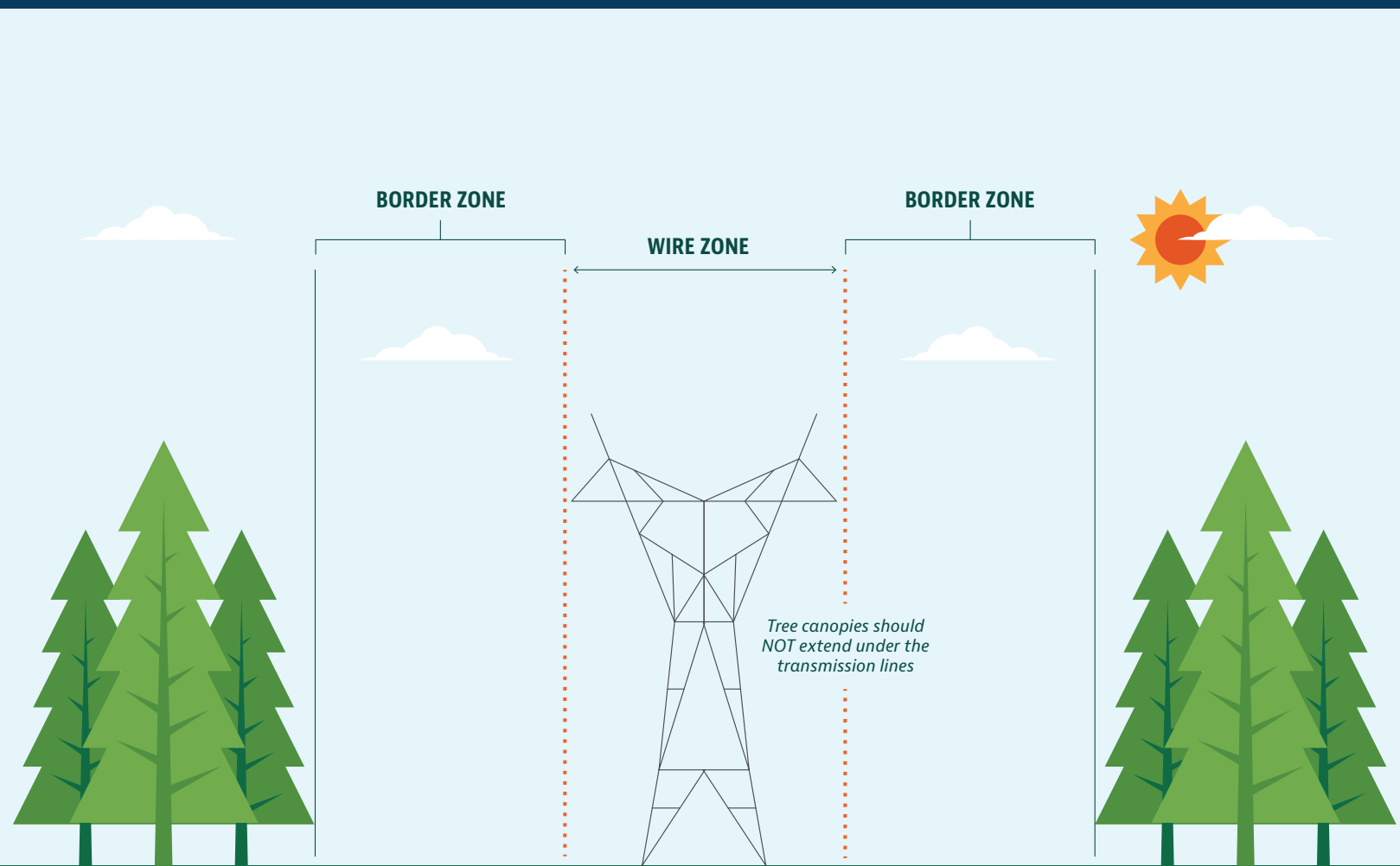
Carry the electric current.

Insulators:

Prevent electricity from grounding.



What Are Transmission Lines?



For safety and reliability, only low-growing vegetation is permitted near transmission lines. Please contact us for more information on maintaining safe distances from our infrastructure.

Rights-of-Way:

Cleared land for safety and maintenance. A right-of-way (ROW) is a designated strip of land that allows transmission lines to pass safely across properties. These corridors are kept free of trees and structures to protect the lines, ensure reliable operation, and provide access for maintenance and emergency repairs. Rights-of-way also help minimize risks to the public and the environment by maintaining a safe distance between high-voltage equipment and nearby areas.

Planning and Building

The process of planning and building transmission projects involves several key stages, each critical to the successful completion of the project:

Planning and Forecasting:

This initial stage involves identifying the need for a transmission project. It includes assessing future energy demands and determining the most efficient and effective ways to meet those needs.

Route selection:

This stage identifies the most efficient and effective path for the transmission lines. This step includes assessing various factors such as environmental impact, land use, and community concerns to determine the optimal route for the project.

Surveying:

Detailed surveys are conducted to gather information about the land, including topography, property boundaries, existing infrastructure, and environmental conditions.

Engineering Design & Planning:

Once projects are identified, the next steps involve developing detailed engineering designs. This includes collaborating with cross-functional teams to ensure design specifications meet safety standards before moving into project execution.

Land Acquisition:

Once the need is identified, the next step is acquiring the necessary land. This involves engaging with landowners and securing the rights to use land for transmission projects.

Soil Testing/

Environmental Research:

Soil testing and environmental research are carried out to ensure the land is suitable for construction and to identify any potential environmental impacts.

Tree-Clearing:

If the land is heavily wooded, tree-clearing may be necessary to prepare the site for construction.

Community Engagement:

Engaging with the local community is crucial to address any concerns and help ensure that the project has the support of those who will be affected by it.

Environmental Review:

An environmental review is conducted to assess the potential impacts of the project and to develop strategies to mitigate any negative effects.

Construction and Commissioning:

This stage involves the actual construction of the transmission infrastructure which includes several sub-stages.

Construction & Commissioning Sub-Stages:

GRADING/FOUNDATION:

Preparing the land and laying the foundation for the transmission structures.

STRUCTURE BUILDING:

Erecting the transmission towers or poles.

TESTING:

Conducting tests to ensure the infrastructure is functioning correctly.

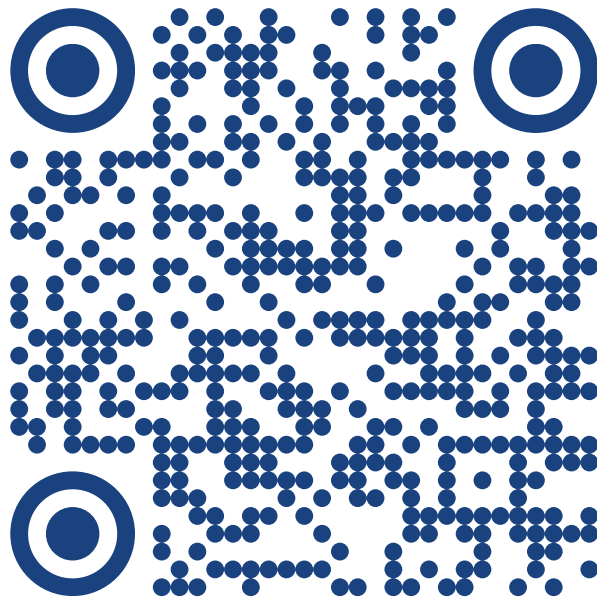
WIRE STRINGING:

Installing the transmission lines.

VEGETATION MANAGEMENT:

Managing vegetation around transmission lines to prevent interference and ensure safety.

**For more information visit
www.georgiapower.com/power
or scan the qr code below.**



**Georgia
Power**